

Juan David Muñoz-Bolaños

Application Engineer Image Processing

Rechenauerstraße 42, Innsbruck, Österreich (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at | [LinkedIn Profile](#)



Summary

Dedicated R&D Optical Engineer with a robust background in combining **High-Precision Optical Metrology** with **Industrial Machine Vision** systems. Expert in developing advanced illumination concepts, real-time image processing algorithms (HALCON/Python), and hardware-software integration for automated inspection units. Proven problem-solving skills with a track record of bridging the gap between complex optical research and reliable industrial deployment.

Professional Experience

INTECOL S.A.S.

Junior Machine Vision System Engineer

Medellín, Colombia

Mar 2018 – Mar 2019

Vision Pipeline Engineering: Designed and implemented Halcon-based beverage bottle inspection systems; developed custom illumination strategies to reliably detect liquid levels and printing defects.

Industrial Integration: Commissioned vision systems using smart cameras and custom illumination for auto-part recognition at a car subdivision of Renault; integrated vision sensors with robotic arms for flame-surface treatment processes.

QA & Feasibility: Performed systematic Gage R&R assessments and feasibility studies for new industrial clients, ensuring 24/7 reliability of vision components on the floor.

Medical University of Innsbruck

PhD Researcher – Optical Metrology & Automation

Innsbruck, Austria

Jan 2022 – Present

System Build & Justage: Led the end-to-end assembly of a custom multiphoton microscope; adjusted galvo scanners, laser sources, and PMTs to achieved sub-micron resolution.

Adaptive Optics Control: Programmed a Python/C++ control stack to synchronize scientific cameras with active wavefront correctors for real-time sample characterization.

Vendor & Component QM: Managed technical specification sheets and qualified precision hardware from Thorlabs and Edmund Optics as part of system procurement.

Leibniz Institute of Photonic Technology (IPHT)

R&D Researcher (Photonics & Data)

Jena, Germany

Jun 2020 – Dec 2021

Instrumentation Design: Prototyping of a dispersed fiber-probe Raman spectrometer; handled optical layout justage and hardware-software calibration.

Data Evaluation Hooks: Integrated computational data analysis (Python ML) with hardware sensors to optimize measurement throughput and validation.

Technical Competencies

Machine Vision Hardware: Mastering of CMOS/CCD camera integration, lens selection (telecentric, macro), and custom illumination design (Dark-field, Bright-field, coaxial).

Image Processing Libraries: Advanced proficiency in Halcon and Cognex VisionPro for real-time inspection, part recognition, and measurement. Familiar with OpenCV and Python-based vision pipelines.

System Alignment & Justage: Hands-on experience in the precise alignment of complex opto-mechanical setups (Laser sources, scanners, detectors). Expert in qualifying components from market-leading vendors.

Automation & Integration: Skilled in automating optical systems with Python, C++, and LabVIEW.

Experienced in ensuring Gage R&R and measurement reliability in harsh industrial environments.

R&D Methodology: Experience bringing systems from concept to series prototypes, including ZEMAX modeling, procurement management, and technical documentation.

Education

Medical University of Innsbruck
Ph.D. in Adaptive Optics & Microscopy

Austria
Jan 2022 – Present

Friedrich Schiller University Jena
M.Sc. in Photonics

Germany
2019 – 2021

University of Cauca
B.Sc. in Physics Engineering

Colombia
2012 – 2018

Publications & awards

Awards: ASML & ZEISS Summer Schools participant; COLFUTURO Grant 2026 awardee.

Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. ACS Photonics 12(1), 176–184 (2025).

Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. Applied Sciences, 14(11), 4726 (2024).

Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. Biomedical Optics Express 14(4), 1562–1578 (2023).

Languages

English (Professional),
German (B2 - Working Proficiency),
Spanish (Native)

Soft skills

Collaboration & Communication: Proven ability to work in cross-functional R&D teams and communicate complex technical concepts effectively.

Problem Solving: Strong analytical mindset with a track record of troubleshooting and resolving hardware-software integration challenges.

Adaptability: Quick learner, highly adaptable to new technologies, and experienced in international industry environments.

Hobbies

Mountain Biking, Dancing Bachata, Learning Harmonica.

References

Prof. Monika Ritsch-Marte
Head of Institute of Biomedical Physics
monika.ritsch-marte@i-med.ac.at

Medical University of Innsbruck

Dr. Christoph Krafft
Spectroscopy Group Leader
christoph.krafft@leibniz-ipht.de

Leibniz Institute of Photonic Technology